

# **Regime Detection in U.S. Equity and Credit Markets Using a Gaussian Hidden Markov Model**

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## Introduction

This learning project is a 3-state Gaussian Hidden Markov Model (HMM) implementation with the objective of identifying market regimes and producing probabilistic forecasts of the regime state (a predictive distribution over the three regimes rather than a single predicted regime). The three latent states represent distinct market regimes: Calm, Transition, and Crisis.

## Dataset Description

Market data is sourced from Yahoo Finance, specifically S&P 500 price and volume. Credit market data is sourced from FRED (Federal Reserve Economic Data). The project originally used the ICE BofA option-adjusted spread series [ high-yield OAS (BAMLH0A0HYM2) and investment-grade OAS (BAMLC0A0CM) ] but as of April 2026 FRED distributes these on a rolling three-year window only, which capped the sample at 731 observations starting in 2023 and left it with no crisis period (neither 2008 nor 2020). The credit series was therefore switched to Moody's seasoned corporate bond yields, Baa (DBAA) and Aaa (DAAA), which are not subject to that restriction and run daily from 1986. Volume is retrieved but is not used as a model input.

Before fitting, the raw data is transformed into three preprocessed input features: log returns of the S&P 500, rolling realized volatility over a 21-day window, and a credit-stress differential. With the new series this differential is Baa minus Aaa rather than high-yield minus investment-grade OAS. Together the three features capture the return, volatility, and credit-stress dimensions. Because the credit series no longer binds the sample, the start date was moved back to 2000, so the data now reaches through the dot-com crash, 2008, and 2020. After feature construction the modeling sample is 6,673 observations, from 2000-02-02 to 2026-08-14 (about 26.6 years).

The two sources use different calendars and different index conventions (yfinance returns a timezone-aware index, fredapi a naive one), so both are normalized before joining. The credit series is forward-filled and prices are not: a yield is a level that still had a value on an unpublished day, whereas a return is a change that does not exist on a non-trading day, and filling it would fabricate a zero return.

## Methods and Tools

The Expectation-Maximization (EM) [Baum–Welch] algorithm estimates the model's parameters; meaning that the initial state distribution, the transition matrix, and each state's emission mean and covariance are estimated by maximum likelihood. The difficulty it addresses is that the regime occupied at each date is latent. If the regime sequence were known, estimation would be direct: each state's mean and covariance would come from the observations assigned to it, and the transition probabilities from the observed frequency of moving between states. But that assignment is exactly what is unknown. EM resolves this circularity by alternating two steps, each conditioned on the current best estimate of the other, and repeating until the observed-data log-likelihood stops improving. Each iteration is guaranteed to not decrease that likelihood, so the procedure converges to a local optimum, which depends on the random starting values and is why the fit is repeated from 15 initializations with the highest-likelihood model kept.

In the E-step, the current parameters are held fixed and the forward–backward recursions compute, for every date, the posterior probability of occupying each state given the observed sequence. This is a soft assignment: rather than committing each observation to a single regime, it distributes it across the three in proportion to how well each explains the data at that point.

In the M-step, those posteriors are held fixed and the parameters are re-estimated as the values that maximize the likelihood under this soft assignment. Each state's mean and covariance become posterior-weighted averages over all observations, and each transition probability becomes the posterior-weighted frequency of the corresponding state-to-state move.

Features are z-score standardized before fitting, since the three differ in scale and the Gaussian likelihood is scale-sensitive. State indices are arbitrary, so states are labeled after fitting by ranking their mean realized volatility: lowest is Calm, highest is Crisis.

Forecasts are produced by propagating the filtered state through the transition matrix, which requires no future observations. This is implemented as an explicit forward recursion rather than `predict_proba`, which returns the smoothed posterior conditioned on the whole sample.

For descriptive purposes, the single most likely regime sequence over the whole sample is recovered using the Viterbi algorithm, which finds the highest-probability path of states given all the observations. This is used only to label history; the regime occupancy figures and the regime bands in Figure 1, and not for forecasting, since it conditions on the entire sample, including data after each date.

## Results and Visualisations

Table 1 : Emission means

|            | log_return | realized_vol | credit_stress |
|------------|------------|--------------|---------------|
| Calm       | 0.0005     | 0.1049       | 0.8149        |
| Transition | 0.0000     | 0.2045       | 1.1057        |
| Crisis     | -0.0006    | 0.4832       | 2.1176        |

**Table 1:** The average value of each feature within each regime. Realized volatility and credit stress both rise from Calm to Crisis, confirming the states are ordered by market stress; the Crisis row also carries the only negative mean return, marking it as a drawdown state.

Table 2 : Transition matrix and expected duration

|            | Calm   | Transition | Crisis | duration_days |
|------------|--------|------------|--------|---------------|
| Calm       | 0.9906 | 0.0094     | 0.0000 | 105.97        |
| Transition | 0.0138 | 0.9839     | 0.0023 | 61.98         |
| Crisis     | 0.0000 | 0.0181     | 0.9819 | 55.25         |

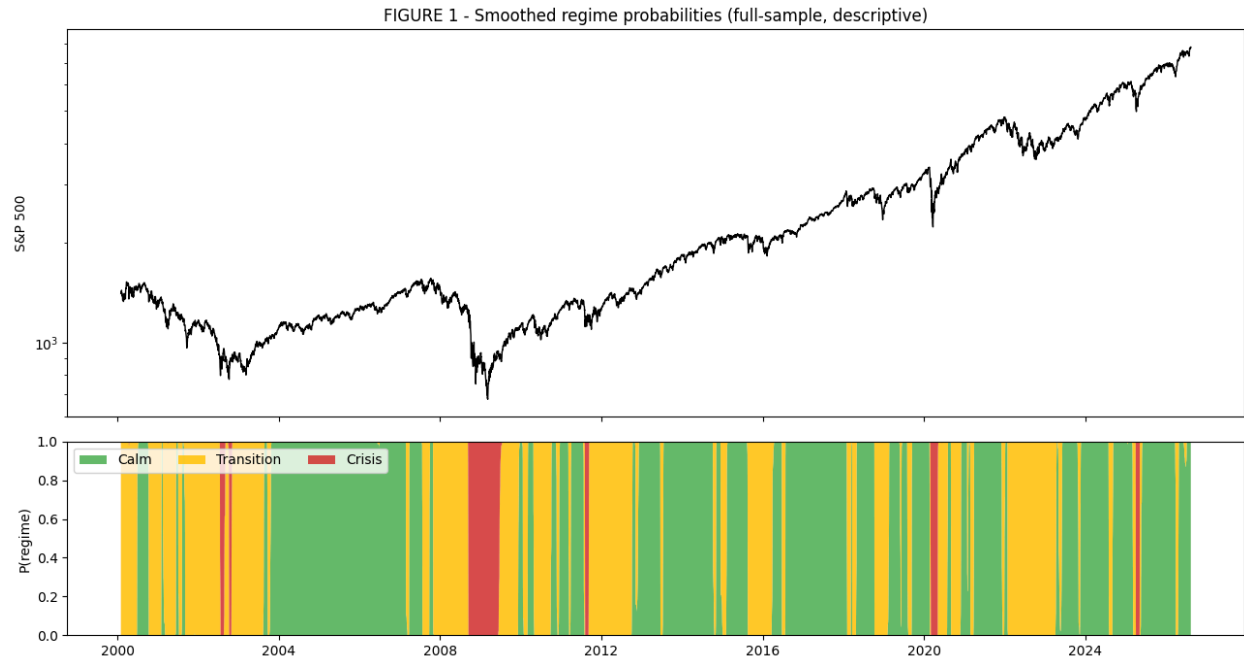
**Table 2:** The probability of moving from each regime (row) to each other (column) in one day, with the expected duration days in the final column (trading days total 252). The high diagonal values show all three states are persistent, and the zero entries between *calm* and *crisis* show the chain cannot jump between the extremes without passing through the *transition* state.

Table 3 : Diagnostics check

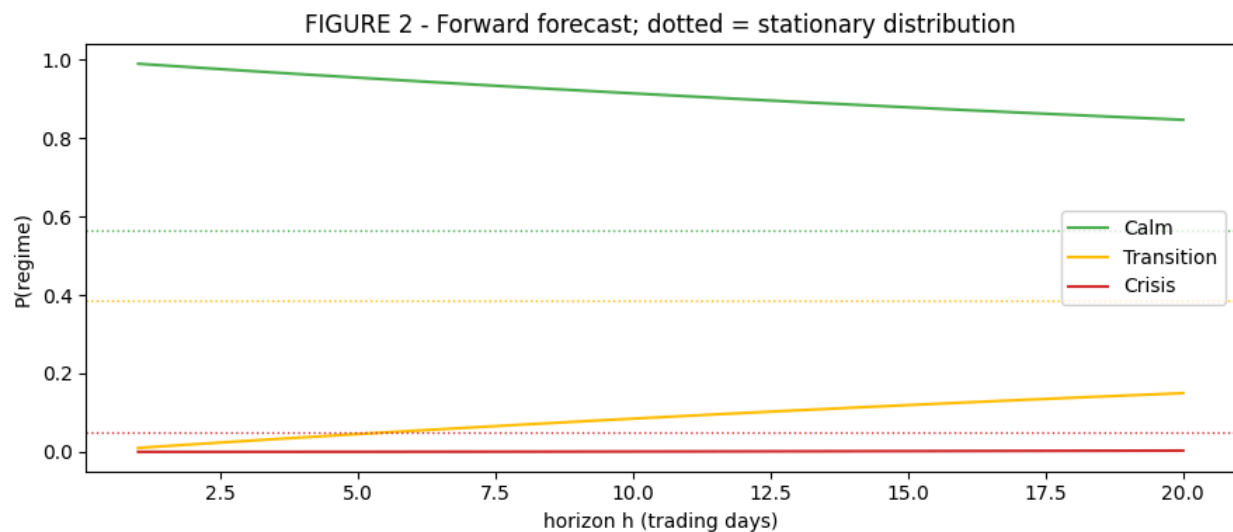
|            | self-persistence | inflow |
|------------|------------------|--------|
| Calm       | 0.9906           | 0.0138 |
| Transition | 0.9839           | 0.0275 |
| Crisis     | 0.9819           | 0.0023 |

**Table 3 :** Diagnostics checked each state for low self-persistence or near-zero inflow; nothing triggered.

Regime occupancy (Viterbi): Calm 56.1%, Transition 38.8%, Crisis 5.1%. The market spends most of the sample in the Calm state and only about 5% in Crisis, matching the expectation that calm conditions dominate and crises are rare. That 5.1% is roughly 340 days, drawn mainly from 2008 and 2020.



**Figure 1 :** The S&P 500 price with the smoothed regime probabilities shown as a stacked band beneath it. Because it conditions on the whole sample, it is a descriptive account of which regime the market occupied over time, not a forecast; the Crisis band lines up with 2008 and 2020.



**Figure 2 :** The predicted regime distribution over the next several days, projected from the final date, against the model's long-run stationary distribution. It shows how quickly the forecast decays from the current state toward the baseline, illustrating the model's short forecast horizon

## Evaluation

Walk-forward one-step-ahead forecasts are generated over the full sample. At each date  $t$  the model, the standardizer, and the filtered state are all derived from data up to and including  $t$  only, and the forecast for  $t+1$  is  $\pi_{t+1} = \pi_t A$ . The model is refit periodically (every 63 trading days), and is frozen between refits. This produced 5,672 one-step-ahead forecasts spanning 2004-01-29 to 2026-08-14.

The choice of scoring target is the following. Two labels are used and reported side by side:

- Causal decode: the regime at each date taken from the filtered posterior using data up to  $t$  only. The forecast is being scored against the model's own decoded state.
- Exogenous volatility tercile: each date is assigned to a low, medium, or high bucket of realized volatility, with cutoffs fixed on the burn-in window so no future information enters. This target is independent of the HMM's transition structure and breaks the circularity of scoring a model against itself.

Two baselines are included because the classes are dominated by calm state; so a raw score is driven mostly by easy days. Climatology predicts the training-period base rates every day. The naive volatility-threshold rule puts together the current realized volatility by the same cutoffs and emits a smoothed one-hot vector.

### Log Score Evaluation

The multi-class log score is the negative log of the probability assigned to the realized class, averaged over all forecasts; lower is better, and it penalizes confident errors very heavily.

| Model         | causal | vol-tercile |
|---------------|--------|-------------|
| HMM           | 0.0493 | 1.3435      |
| Climatology   | 0.8773 | 1.0033      |
| Vol-threshold | 0.7372 | 0.1054      |

**Table 4:** The multi-class log score for the model and both baselines, under each label.

Against the causal label the HMM scores far below both baselines. Against the exogenous label it scores worse than climatology, and the volatility-threshold baseline scores best; but that baseline is built from the same quantity the label is built from, so it is effectively reading the answer key rather than forecasting.

## Brier Score Evaluation

The multi-class Brier score is the summed squared error between the forecast vector and the one-hot realized class, averaged over all forecasts; lower is better, and it is softer on overconfidence than the log score.

| Model         | causal | vol-tercile |
|---------------|--------|-------------|
| HMM           | 0.0221 | 0.4277      |
| Climatology   | 0.5497 | 0.5938      |
| Vol-threshold | 0.3867 | 0.0150      |

**Table 5:** Brier score.

The pattern is the same under the causal label. Under the exogenous label, however, the HMM now beats climatology (0.4277 vs 0.5938), where on the log score it had lost. This disagreement between the two scores is informative and is discussed below.

## Combined Evaluation

| Model         | log (causal) | brier (causal) | log (vol) | brier (vol) |
|---------------|--------------|----------------|-----------|-------------|
| HMM           | 0.0493       | 0.0221         | 1.3435    | 0.4277      |
| Climatology   | 0.8773       | 0.5497         | 1.0033    | 0.5938      |
| Vol-threshold | 0.7372       | 0.3867         | 0.1054    | 0.0150      |

**Table 6:** Combined. Both scores under both labels in a single view, so the model and baselines can be compared across all four columns at once. It consolidates Tables 4 and 5 and makes the log-versus-Brier split directly visible on one row.

## Interpretation of Evaluations

Under the causal decode the HMM beats both baselines, but this mostly shows internal consistency rather than model accuracy: the causal label is the model's own filtered state, so it only checks whether each forecast agrees with the next day's decoded state.

The volatility-tercile label is the more independent test. The volatility-threshold baseline scores best under it, but only because both the baseline and the label are built from realized volatility, so it matches the target by construction rather than forecasting. The fairer comparison is the HMM against climatology, and here the scores disagree: the HMM beats climatology on Brier (0.4277 vs 0.5938) but loses on the log score (1.3435 vs 1.0033).

This disagreement is the main result of my project. Brier rewards putting probability roughly in the right place; the log score punishes being confident and wrong. Doing better on Brier but worse on the log score points to overconfidence; therefore based on this the model as some good indicator of regime direction but is nonetheless overconfident. In other words, it has some predictive value but it is far from a certain prediction.

### **Analysis and Conclusion**

Four features of the fitted model stand out, these were observed in the smaller original sample and are strengthened by the move to the full 26-year sample.

First, the regime structure forms a ladder. The transition matrix cannot jump between Calm and Crisis in either direction; both entries are estimated at exactly zero, so the chain must pass through the Transition state to move between the extremes. This was not imposed: the transition matrix was estimated with all entries free, so the zeros emerged from the data, and they hold on the full sample rather than only the earlier short window.

Second, the regimes are persistent and seemingly reasonably ordered in duration. Calm persists longest at about 106 trading days (roughly five months), Transition about 62, and Crisis about 55. Turbulent states are shorter-lived than quiet ones, which matches the intuition that crises are sharp and calm states are more persistent.

Third, the Crisis state corresponds to genuine crises. At 5.1% occupancy it is estimated from roughly 340 observations, drawn mainly from 2008 and 2020, rather than the roughly 25 observations of the earlier three-year window. Its mean return is negative (-0.0006), consistent with real drawdowns, and its inflow remains the lowest of the three states at 0.0023, as expected for the regime the market enters least often.

Fourth, the forecasts carry real but limited information. As the evaluation showed, the model beats climatology on the Brier score but not the log score, a split that points to overconfidence: it indicates regime direction reasonably well but states it with overcertainty.



## Challenges

- Timezone mismatch: yfinance returns a timezone-aware index and fredapi a naive one; joining directly yields a near-empty result with no error. Both are now normalized first.
- API key handling: I initially wrote the key into a notebook cell and so it would have been saved and submitted with it. Since I want to share on github, I moved it to Colab's secret manager. Because getpass hides input, a mispaste then went unnoticed until it appeared as an unrelated-looking URL error, so I added key validation.
- EM local optima: Different seeds gave materially different fits; resolved with 15 restarts keeping the best likelihood.
- Credit series availability: FRED restricted the ICE BofA OAS series to a rolling three-year window (April 2026), which capped the sample at 731 rows from 2023 and left it with no crisis period. Resolved by switching to Moody's Baa/Aaa yields, as described under Dataset Description. Because both the feature definition and the sample changed, the earlier ICE-based results and the current results are not directly comparable.
- Overarching: Since I used the help of AI tools, I felt at times as though I was overreaching in parts of this project. As such, I tried to learn these parts as I went, using the methods as I work on my understanding of them. This did lead to learning things out of order given my current academic background, which was at times uncomfortable, but it is a gap I intend to close properly as my coursework catches up in a master's and its bridging coursework.

## References

L. E. Baum, T. Petrie, G. Soules, and N. Weiss, "A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains," *Ann. Math. Statist.*, vol. 41, no. 1, pp. 164–171, 1970.

## Data

S&P Dow Jones Indices, "S&P 500 (^GSPC)," Yahoo Finance. [Online]. Available: <https://finance.yahoo.com/quote/%5EGSPC>

Moody's, "Moody's Seasoned Baa Corporate Bond Yield (DBAA)," FRED, Federal Reserve Bank of St. Louis. [Online]. Available: <https://fred.stlouisfed.org/series/DBAA>

Moody's, "Moody's Seasoned Aaa Corporate Bond Yield (DAAA)," FRED, Federal Reserve Bank of St. Louis. [Online]. Available: <https://fred.stlouisfed.org/series/DAAA>

Ice Data Indices, LLC, "ICE BofA US High Yield Index Option-Adjusted Spread (BAMLH0A0HYM2)," FRED, Federal Reserve Bank of St. Louis (original series, since superseded). [Online]. Available: <https://fred.stlouisfed.org/series/BAMLH0A0HYM2>

Ice Data Indices, LLC, "ICE BofA US Corporate Index Option-Adjusted Spread (BAMLC0A0CM)," FRED, Federal Reserve Bank of St. Louis (original series, since superseded). [Online]. Available: <https://fred.stlouisfed.org/series/BAMLC0A0CM>

## Software

hmmlearn, yfinance, fredapi, pandas, NumPy, scikit-learn, SciPy, and Matplotlib (Python libraries).

## AI Resources

These resources were used in review and debugging of code, as well as for searching for pertinent information and sources. OpenAI's ChatGPT, Anthropic's Claude.